

Chapter 8

- 8.1. It is known that a RLC circuit is shown in Figure 8-1. The input voltage is $v_i(t)$, and the output of the circuit is $v_C(t)$ at both ends of the capacitor. Try to write the equation of state and the output equation of the circuit

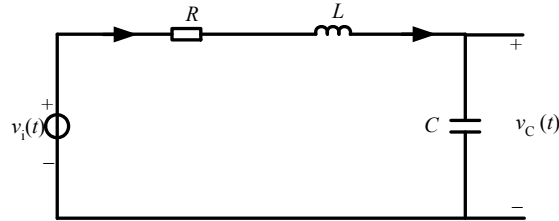


Figure 8-1

- 8.2. The system function of a continuous LTI system is

$$H(s) = \frac{50s}{s^2 + 3s + 1}$$

- (1) Write the differential equation of the system
 - (2) Draw out the direct simulation block diagram of the system
 - (3) Write the state equation and output equation of the system by simulation block diagram.
- 8.3. For the simulation block diagram of each system shown in Figure 8-2, we try to write out the state equation and output equation of the system.

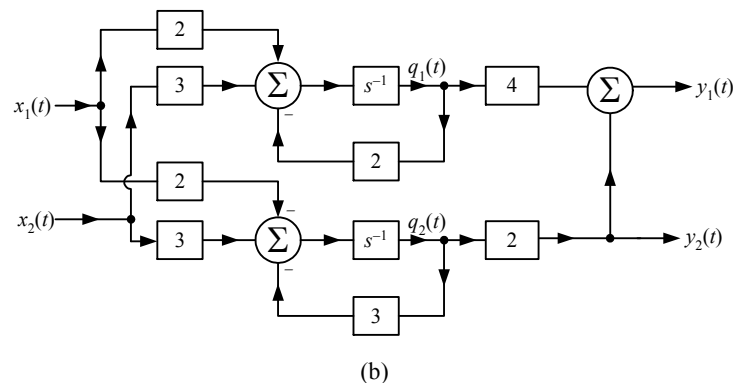
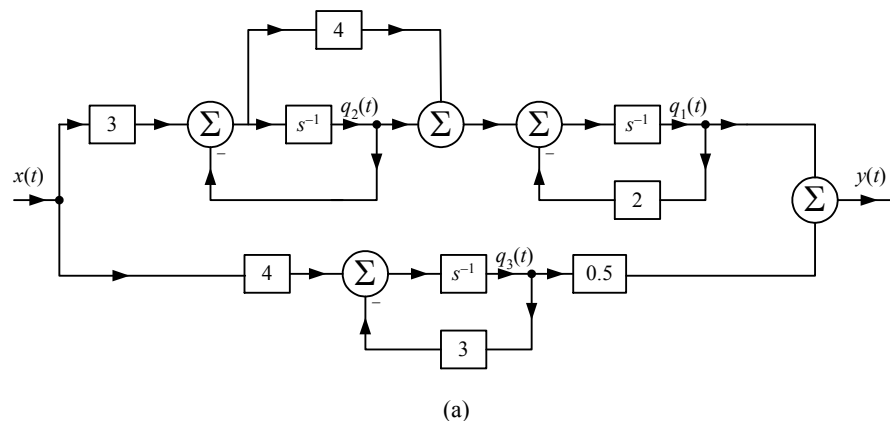


Figure.8-2

- 8.4. We know the system function of a discrete LTI system is

$$H(z) = \frac{2z^2 + 3}{z^2 - 1.96z + 0.8}$$

- (1) Draw system simulation block diagram
- (2) Write out the state equation and output equation of the system by simulation block diagram.
- 8.5. It is known that the simulation block diagram of a discrete LTI system is shown in figure 8-3, and write out the state equation and output equation of the system.

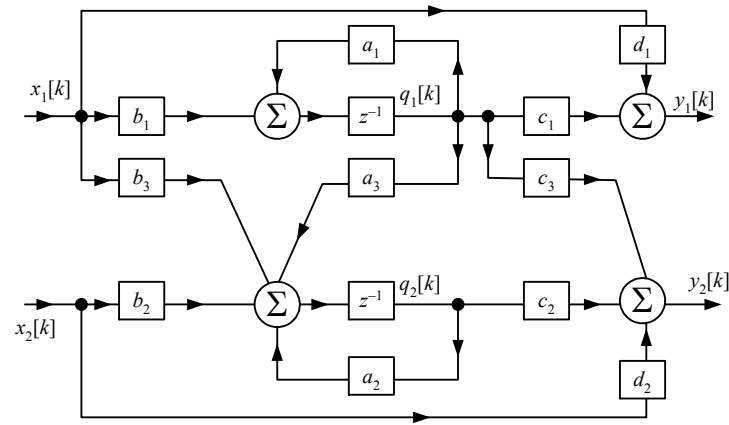


Figure.8-3